

Anion-Mediated Allosteric Catalysis of [2 + 2] Photocycloaddition Based on a Flexible Metallo-Amine Cage for High Diastereoselectivity

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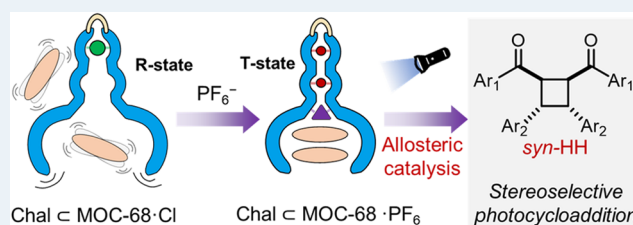
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ABSTRACT: Anions can exert a notable influence on the solubility and folding of protein structures, consequently altering their activity and stability due to the well-known Hofmeister effect. Herein, we report a protein-like allosteric metal–organic cage (MOC-68) with the solubility and conformation regulated through anion exchange, leading to conversion between the relaxed state (R-state) and the tensed state (T-state) to show distinct host–guest interactions. This conformational change enables allosteric catalysis of intermolecular [2 + 2] photocycloaddition of acyclic olefins with greatly enhanced yield and diastereoselectivity. Kinetic experiments reveal that the T-state of MOC-68 is superior in catalytic performance for the steric and thermodynamic disfavored *syn*-HH product, displaying a threefold rate enhancement over the R-state and totally a sevenfold increase owing to the reaction acceleration contributed by an enzymatic cage effect. Notably, this allosteric cage exhibits exceptional stability and recyclability, maintaining consistent catalytic efficiency and selectivity over five cycles. This work highlights the potential of structural allostery in flexible cages to enhance the substrate-binding ability and catalytic selectivity, providing a way to advance enzyme-mimicking cage catalysis to regulatable and efficient allosteric catalysis.

KEYWORDS: metal–organic cage, allosteric catalysis, [2 + 2] photocycloaddition, anion exchange, diastereoselectivity



INTRODUCTION

Protein allostery plays a pivotal role in the regulation of biological processes, underpinning a fundamental mechanism for modulating protein and enzyme functions in nature.^{1,2} Through the binding of a preceding effector, such as anion, cation, or a small molecule, conformational change occurs to alter the affinity of the subsequent binding behaviors, which could further impact the function of the biomacromolecule, e.g., changing its biological activity or catalytic selectivity.^{1,3,4} This phenomenon has inspired researchers to explore the potential of structural allostery in design and development of advanced supramolecular materials and devices, including metal–organic cages (MOCs),^{5–11} organic cages,^{12–14} molecular machines,^{15,16} etc., to achieve improved guest capture and catalytic performance.^{17–23}

As a paradigm of artificial receptor and reactor, MOCs feature in the tailored cage structures, which are able to integrate varied physicochemical properties into the confined nanospace, enrich host–guest interactions, and create micro-heterogeneity in a cage solution,^{24–26} thus motivating unique applications in guest recognition,^{27–30} separation,^{31–35} and catalysis.^{26,36–41} The well-organized MOCs with the confined cavities can offer new possibilities for chemical reactions, such as the reactions' acceleration,^{25,26,42} the regulation of

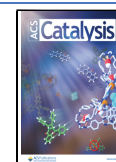
selectivity,^{43–45} and, more importantly, the supramolecular catalysis for challenging organic transformations by integrating functional groups into the cage.^{40,46} This allows the artificial cage reactors not only to emulate protein characteristics but also to exhibit reactivities that can rival or even surpass the enzymes. In most cases, MOCs constructed from rigid ligands lack the flexibility to undertake allosteric or adaptive behaviors without altering the coordination environment of metal centers. Therefore, investigations mainly focus on the well-shaped cavities, which serve as “molecular flasks” to achieve specific catalytic purposes.^{47,48} Alternatively, flexible MOCs are able to endow catalysis with the turn-on and turn-off behaviors, or the activity control through allostery and induced-fit processes.^{49–54} Nonetheless, it still remains a challenge to regulate the selectivity and enhance the performance of cage catalysis artificially by virtue of allosteric strategy, particularly

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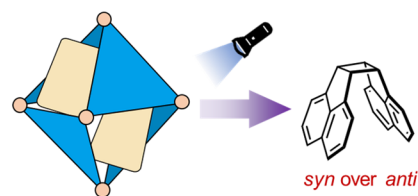
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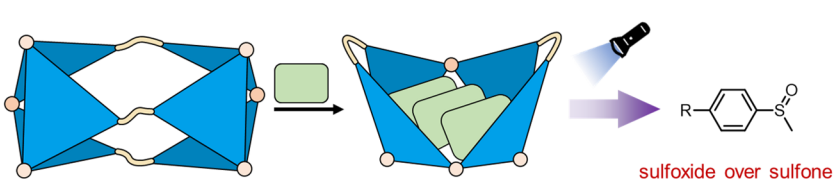
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Previous work: Rigid cage flask



Induced-fit cage



This work: Artificial allosteric cage for selective photocatalysis

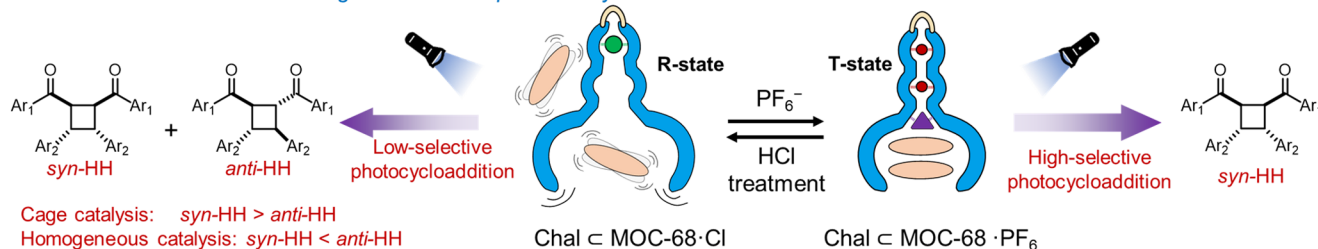


Figure 1. Representative photoreactions in a rigid cage flask,⁴⁷ induced-fit cage,⁵² and the artificial allosteric cage.¹¹ The colored rectangles and ellipses represent the guest substrates. The allosteric tetrahedral cage in this work was simplified as a clip-shaped model, demonstrating one of the six pockets of MOC-68 to show the allosteric sites and the guest-binding sites. The colored balls denote the effectors of Cl[−] anion (green) and water molecule (red) located inside the apical amine tripods, the purple triangle denotes the coeffector of PF₆[−] anion in the central cavity, and Chal refers to the chalcone guest substrate.

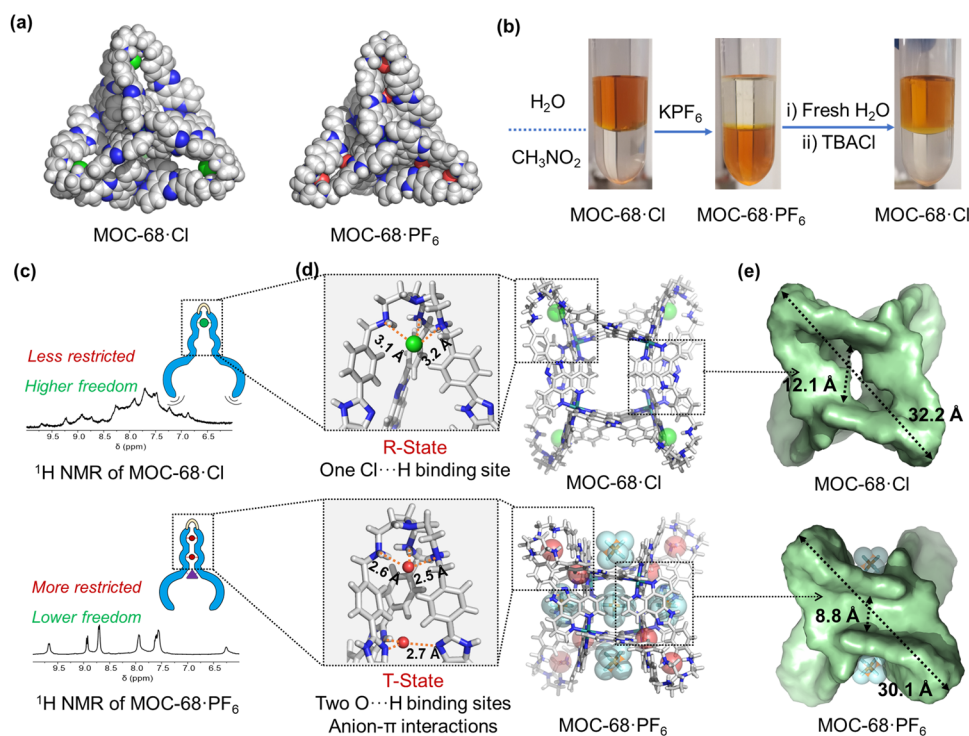


Figure 2. Anion-mediated allosteric effect on MOC-68. (a) Cage structures showing expanded and shrunken tetrahedral shape with Cl[−] (green ball) or H₂O (red ball) effectors binding in the tris-amine and tris-imidazole endocavities of MOC-68·Cl and MOC-68·PF₆, respectively. (b) Anion-regulated solubility change and structural conversion via a phase transfer process. (c) ¹H NMR resonance change from the less-restricted MOC-68·Cl to the more-restricted MOC-68·PF₆. (d) Crystal structures of MOC-68·Cl and MOC-68·PF₆ showing anion positions and highlighting the conformational changes driven by effectors' and coeffectors' binding. (e) Surface visualization of MOC-68·Cl and MOC-68·PF₆ showing the overall cage size and guest-accessible pocket size.

in the recently renaissance area of asymmetric photocatalysis (Figure 1).^{55–57}

We previously reported an allosteric metallo-amine cage (MOC-68) that contains four flexible amine tripods on its tetrahedral vertices.¹¹ An interesting feature of this flexible cage is that the conformational change can be regulated by acid and

base with water molecules as the effectors, leading to a control of the multimode host–guest dynamics. Herein we demonstrate the anion-mediated allostery of this cage, which enables the distinct host–guest interacting behaviors and further remarkably influences the catalytic performance in [2 + 2] photocycloaddition. The anions are omnipresent in both

proteins and the cationic cage solution with a variety of bonding modes,^{58–60} which can also act as effectors or coeffectors to control not only the solubility^{34,61} but also the conformation depending on the Hofmeister effect.^{3,62} We found that the spherical Cl^- anion can replace the water effectors in MOC-68 to result in the less-restricted MOC-68-Cl in R-state (Figure 1), showing weaker host–guest interactions and better water solubility, while the bulky PF_6^- anions can act as coeffectors to trigger conformation tightening in MOC-68- PF_6 to restore the T-state, showing stronger host–guest interactions and better solubility in organic solvents. In contrast to normal homogeneous catalysis without enzymatic cage effect, unconventional *syn*-HH preferential diastereoselectivity of [2 + 2] photocycloaddition of acyclic olefins is achievable by this flexible cage, of which the T-state exhibits much enhanced yield and diastereoselectivity than the R-state. Particularly, the diastereoselectivity has been remarkably improved for those substrates that were previously challenging to attain by using rigid MOC-16,⁴⁴ suggesting a way to realize superior catalytic performance by implementing structural allostery on the flexible cages.

RESULTS AND DISCUSSION

Anion-Directed Cage Conformation Change. MOC-68-Cl and MOC-68- PF_6 containing different counteranions were synthesized according to the modified procedure in our previous work (see the synthesis and characterization details in Supporting Information, Figures S1–S20).¹¹ As confirmed in a previous study, the excellent chemical stability of MOC-68 under harsh conditions and the inert Ru–N coordination of metal centers ensure that the cage retains structural integrity when treated with substantial Cl^- and PF_6^- anions. The high-resolution electrospray ionization mass spectrometry (ESI–MS) verified the cage structure unity in solution by displaying a series of successively charged species that are consistent with the compositions containing Cl^- and PF_6^- anions (Figures S12 and S15), and the single-crystal X-ray diffraction analyses unambiguously revealed the tetrahedral framework of the cage and the location of counteranions and water molecules (Figures 2 and S20).

Both MOC-68-Cl and MOC-68- PF_6 are composed of a cationic cage and counteranions that balance positive charges. The cationic cage comprises 4 Ru-metalloligands and 4 amine tripods, showing a tetrahedral shape with 4 same screwed Ru-metalloligands occupying the faces, thus imparting homochirality to the cage.¹¹ However, distinct anion-cage interactions are found, which makes the cage conformations noticeably different. In MOC-68-Cl, four spherical Cl^- anions are located in the four tris-amine endocavities on amine tripods, forming N–H...Cl hydrogen bonds (HBs) with three aliphatic –NH-amines. In the solid state, the homochiral cages are connected through the outer Cl^- anions to assemble into a chiral chain with 3-fold screw axis symmetry imposed, spontaneously resolving into chiral crystals in $P6_22$ or $P6_422$ space groups of opposite chirality (Figure S18 and Table S1). In contrast, the larger weakly hydrogen-binding PF_6^- anions are located in the central cavity and four of six open pockets of the cage, forming anion- π and electrostatic interactions with the cage. Notably, the smaller and strongly H-binding water molecules occupy the tris-amine and tris-imidazole endocavities on the four vertices,¹¹ forming O–H...N and N–H...O HBs with three aliphatic –NH- amines and three imidazole-N/NH groups (Figure 2d).

These divergent anion-cage interactions consequently cause conformational changes in MOC-68, which is evident from the torsion angles in the flexible amine tripods (Figure S19). In MOC-68- PF_6 , the angles are generally decreased compared with those in MOC-68-Cl (94.8 vs 84.5°; 128.6 vs 120.6°), leading to contraction of the entire cage and, more significantly, narrowing the six pockets that are formed by the rigid Ru-metalloligands due to a fan-like screwing. As seen from Figure 2e, although the overall cage size of MOC-68 only slightly reduces from 32.2 to 30.1 Å, the pockets compress remarkably from 12.1 to 8.8 Å, corresponding to a tightened form of the T-state in MOC-68- PF_6 and an expanded form of the R-state in MOC-68-Cl. The endocavities and the pockets in R-state are interconnected, presenting a total volume of 614.3 Å³ as calculated by MoloVol (Figure S20).⁶³ In contrast, the six pockets in T-state are well defined to present individual volumes (31.8–48.3 Å³), summing a total volume of up to 256.9 Å³. The above conformational analyses suggest that the anions play a vital role in regulating the microenvironment of the cavities in MOC-68, where Cl^- anions themselves act as effectors in lieu of water molecules to relax the cage conformation, while the PF_6^- anions behave as coeffectors to assist recovery of the tightened cage conformation, leaving small allosteric sites occupied by H-binding water molecules. Therefore, the anion exchange exerts an allosteric effect to mediate the host–guest interactions, shrinking the total accessible cavity volume by ~58% and the pocket size by ~27%, which subsequently impact the guest fit and exchange dynamics.

Anion-Mediated Allostery. Similar to proteins, the solubility of MOC-68 is alterable by counteranions (Table S2).⁶¹ Hydrophilic Cl^- anions impart high water solubility to MOC-68-Cl, which can readily dissolve in water but is completely insoluble in most organic solvents like MeCN, CH_3NO_2 , and DMF. On the contrary, the less hydrophilic PF_6^- anions reverse the solubility preference of MOC-68- PF_6 . This solubility variation allowed us to develop a phase transfer protocol for the anion conversion, implemented through anion exchange between a water solution and an immiscible organic solution (Figure 2b). We initially dissolved MOC-68-Cl in water, which is layered on the CH_3NO_2 solvent. When adding KPF_6 (8 equiv) into the double-layer system, MOC-68 was able to transfer into the lower organic solvent after shaking. By replacing the upper layer with fresh water and adding excess tetrabutylammonium chloride (TBACl, 20 equiv), MOC-68 could be transferred back into the water layer. Delightfully, a CD_3CN - D_2O mix solvent manifests excellent solubility for both MOC-68-Cl and MOC-68- PF_6 . Hence, the ^1H NMR measurements were performed in a $\text{CD}_3\text{CN}:\text{D}_2\text{O} = 1:3$ (v/v) medium to clarify the solution structure (Figures 2c and 3 and S21 and S22).

Similar to the previously studied MOC-68 containing BF_4^- anion, MOC-68- PF_6 exhibits a clear and assignable set of NMR resonances in acidic conditions (Figures S7–S11), indicating a uniform conformation of MOC-68 in the T-state in solution.¹¹ In contrast, the ^1H NMR resonances of MOC-68-Cl are vague and hard to discriminate (Figure S13), suggesting the presence of multiple conformations in solution.^{11,64} These findings are consistent with the expectation that the relaxed MOC-68-Cl may allow a higher degree of motion freedom of the ligands, thus resulting in less-restricted and variable cage conformations to broaden ^1H NMR resonances. In comparison, the PF_6^- anions could stiffen the cage conformations in a

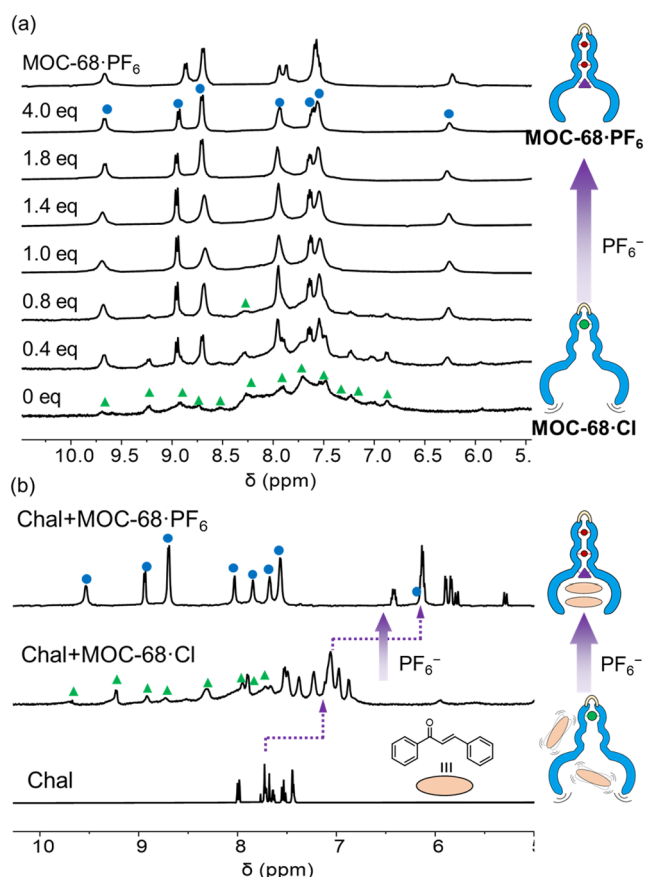


Figure 3. (a) ^1H NMR titration of MOC-68-Cl with KPF_6 . (b) ^1H NMR changes of the MOC-68 cage and Chal guest mediated by anion exchange (400 MHz, 298 K, $\text{CD}_3\text{CN}:\text{D}_2\text{O} = 1:3$, v/v, pD = 3.0). Guest shifts are denoted by dashed purple arrows; proton signals of MOC-68-Cl are marked in green triangles; proton signals of MOC-68- PF_6 are marked by blue dots; and allosteric behaviors are illustrated by clips.

tightened form to unify the proton resonances (Figure 2c). From ^{19}F -NMR, two different types of PF_6^- can be observed (Figures S8 and S9). The minor signal is assigned to the endobinding species, while the major signals correspond to the exchangeable exobinding species. In each case, the ^1H -DOSY experiments confirm the sole existence of MOC-68 structure in solution with all cage proton signals appearing on a line (Figure S16). MOC-68-Cl displays a slightly smaller diffusion constant ($D = 1.38 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$) than that of MOC-68- PF_6^- ($D = 1.74 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$), indicative of a larger hydrodynamic radius of MOC-68-Cl, which is in line with the single-crystal analysis result.

To understand the anion-mediated regulation process, we employed ^1H NMR titration experiments to monitor the cage allosteric behaviors driven by anions (Figures 3 and S21 and S22). Upon aliquot addition of KPF_6 into the MOC-68-Cl solution ($\text{CD}_3\text{CN}:\text{D}_2\text{O} = 1:3$, v/v), the broad cage resonances became sharp and unified, clearly showing a structural conversion process that finally turned out an NMR profile in accordance with that of the prepared MOC-68- PF_6 (Figure 3a). This demonstrates that PF_6^- is capable of driving an R-state-to-T-state transformation from the less-restricted MOC-68-Cl to the more-restricted MOC-68- PF_6 . Intriguingly, this conversion process starts immediately when a small portion of PF_6^- is added, giving an equilibrium between the R-state and

T-state that is completely converted to the T-state after the addition of just 1 equiv KPF_6 , far below the theoretical charge-balanced stoichiometric ratio of the 8+ cation cage (8 equiv). As known from the single-crystal analyses, the relatively bulky and weakly H-binding PF_6^- anions are not fit into the small tris-amine endocavities; instead, they form anion- π and electrostatic interactions with the cage walls to occupy the central cavity and the open pockets of the cage. Furthermore, the MS titration experiments more clearly revealed this transition (Figure S23). When adding 1 equiv of KPF_6 to the MOC-68-Cl solution, the Cl^- binding signal of the cage disappeared, and a series of differently charged signals corresponding to the cage with association of water molecules and one PF_6^- emerged. The presence of up to eight bound water molecules further indicates that the addition of just 1 equiv of PF_6^- could promote the cage conformational transition, subsequently squeezing out Cl^- ions to be replaced by the binding water molecules. These titration results indicate that only one PF_6^- binding is sufficient to trigger conformational conversion completely, suggesting that PF_6^- actually acts as a coeffector to assist restoration of MOC-68 from R-state to the compressed T-state, during which 4 larger Cl^- anions are squeezed out and 8 smaller water molecules are fit in as effectors concurrently (Figures 2d and 3a). We further tested whether a reverse conversion from MOC-68- PF_6 to MOC-68-Cl could happen. As seen from Figure S22, there was no obvious signal change even when an excess of Cl^- was added to the MOC-68- PF_6 solution. This result indicates that PF_6^- has a strong stabilization effect to keep the T-state of MOC-68 in an aqueous mixture of $\text{CD}_3\text{CN}:\text{D}_2\text{O}$. Nevertheless, a transition from the T-state to the R-state can be achieved through a precipitation process by treating the MOC-68- PF_6 solution with HCl and subsequently removing the PF_6^- anions (Figure S24).

To explore how such anion-mediated cage allostery impacts the host-guest interactions, we examined the guest inclusion behaviors of MOC-68-Cl and MOC-68- PF_6 with an aromatic chalcone (Chal) guest (Figure 3b). When adding 4 equiv of Chal into MOC-68-Cl solution, the ^1H NMR signals of Chal show an obvious upfield shift, while those of the cage are only slightly sharpened, suggesting a loose binding of Chal molecules to move proton signals due to the shielding effect of aromatic walls of the cage. Strikingly, when 1 equiv of KPF_6 was added, the guest signals further move upfield, showing ~ 2 ppm shift compared with the free guest. Additionally, the cage signal profile changes completely to be akin to MOC-68- PF_6 completely. From this, it is evident that the addition of PF_6^- effectively induces cage allostery, consequently modulating the host-guest interacting behaviors.

Detailed host-guest interactions of MOC-68- PF_6 and MOC-68-Cl were further investigated by ^1H NMR titration and ESI-HR-MS with Chal guests (Figures 4 and S25–S33). In order to exclude the background interference, we first titrated a blank $\text{CD}_3\text{CN}:\text{D}_2\text{O}$ solution (1:3, v/v) with Chal molecules (Figure S25). It is clear that most Chal proton signals show a minor upfield shift (0.01 ppm) upon continuous increase of Chal concentration from 0 to 6 mM. Interestingly, a new set of Chal proton signals starts to appear and increase when the concentration reaches 2 mM, suggesting aggregation of Chal molecules at this point.

When Chal was proportionally added into the $\text{CD}_3\text{CN}:\text{D}_2\text{O}$ solution (1:3, v/v) of MOC-68- PF_6 , the cage and guest signals displayed diverse shifts (Figures 4 and S26). The guest signals

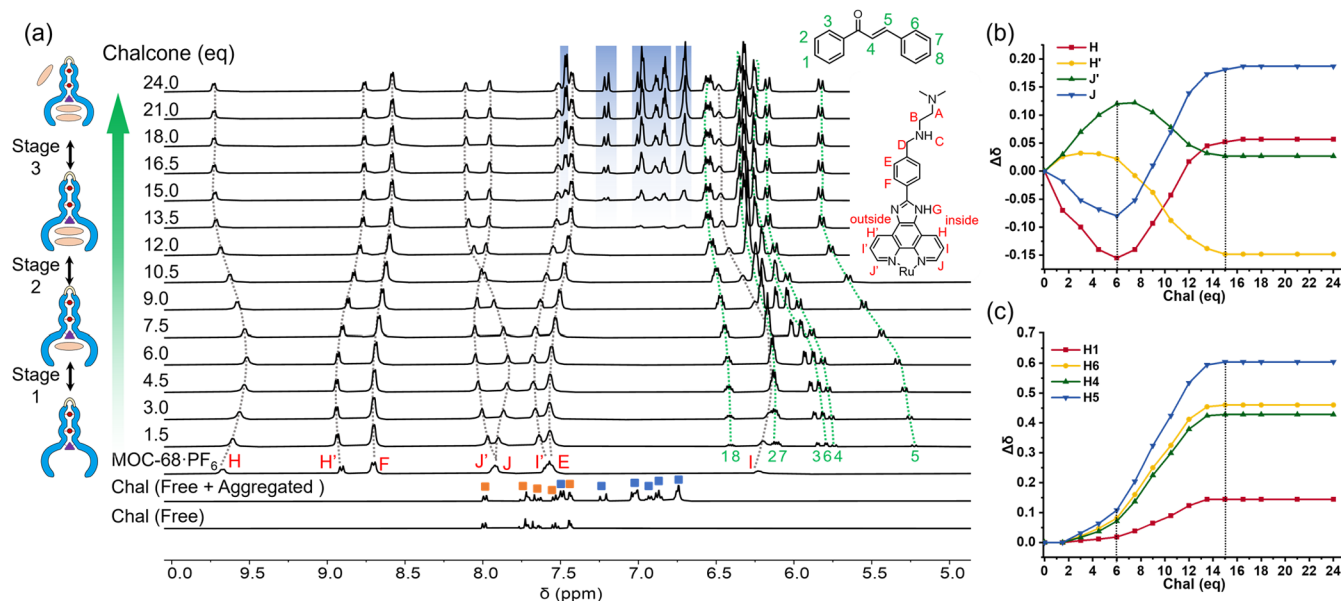


Figure 4. (a) ^1H NMR titration of MOC-68- PF_6 with Chal guests (600 MHz, $\text{pD} = 3.0$, $\text{CD}_3\text{CN}:\text{D}_2\text{O} = 1:3$, v/v, MOC-68- $\text{PF}_6 = 0.5$ mM, 298 K). Free Chal signals are denoted by orange squares, and aggregated Chal signals are denoted by blue squares and highlighted in blue shadow. Signal shifts of MOC-68 and bound Chal are tracked with colored dash lines. (b) ^1H NMR chemical shift curves of representative protons on MOC-68- PF_6 . (c) ^1H NMR chemical shift curves of representative protons on bound Chal. The insets show proton indexing of Chal guest and MOC-68, where proton locations inside or outside the cage are distinguished.

move upfield by more than 2 ppm at the beginning and then shift downfield gradually upon continuous titration. An inflection point of signal moving can be found at 6 equiv Chal, and saturation of guest uptake may be reached until 15 equiv Chal with the maximum signal shift (Figure 4c). At about 13.5 equiv of Chal (equal to 6.75 mM), a new set of signals emerges, which is ascribed to Chal aggregation in bulk solution, as observed in the above blank titration. Considering the dynamic exchange of guest ingress-and-egress that always happens, which is evident since only a single set of cage and guest resonances is observed except for the aggregated guest signals, the NMR integration of the host and guest signals at the Chal saturation point indicates that the maximum guest-binding capacity of MOC-68- PF_6 for Chal can be estimated to be 12, corresponding to two guests in each of the six open pockets (Figures S27). The host–guest ^1H -DOSY and ^1H – ^1H NOESY measurements confirm the close interactions between MOC-68 and Chal (Figures S28 and S29). The DOSY diffusion coefficients of the cage and guest are comparable, and the NOE results indicate that the Chal guests bind in the center of the cage pocket, showing cross peaks of $\text{H}^{\text{H}}\text{--}\text{H}^{\text{S}}$ (phenanthroline H on the cage and Chal H on the middle double bond) and $\text{H}^{\text{F}}\text{--}\text{H}^{\text{B}}/\text{H}^{\text{F}}\text{--}\text{H}^{\text{I}}$ (phenyl H on the cage and Chal Hs on the two sides). The ESI–HR-MS detection of the ChalMOC-68- PF_6 complex also suggested the association of Chal guests with the cage in solution (Figures S32). Based on these spectroscopic results and the single-crystal structure, a guest-binding model of ChalMOC-68- PF_6 has been simulated (Figures S33), where each elongated pocket is suitable for the dual stacking of two Chal molecules.

The cage signal shifts during the titration process are more informative and can be divided into three distinct stages (Figure 4a). In the initial stage (0–6 equiv of Chal), the addition of Chal induces upfield signal shifts of H^{H} , H^{I} , and H^{I} that locate on the phenanthroline motif facing the inside of the cage, whereas the outside partners, $\text{H}^{\text{H}'}$, $\text{H}^{\text{I}'}$, and $\text{H}^{\text{I}'}$, move

downfield. This stage is attributed to the first guest binding in each of the 6 open pockets, well in agreement with the inflection point of the guest signal moving at 6 equiv Chal. When more Chal was added, these cage peaks shifted backward, up to a plateau, with 6 open pockets fully occupied by the second guest (12 equiv Chal). This second stage continues until 15 equiv of Chal (Figure 4b), considering the fact that more than 12 equiv of Chal is needed to counteract the guest exchange into bulk solution. Afterward, additional Chal could no longer induce the signal shift; instead, the Chal molecules tended to aggregate in bulk solution or precipitate from the solution. These distinct movements of the cage signals suggest stepwise binding of MOC-68- PF_6 to the Chal guests.

In comparison, the titration of MOC-68-Cl in $\text{CD}_3\text{CN}\text{--}\text{D}_2\text{O}$ solution (1:3, v/v) with Chal suggests much weaker host–guest interactions, which can be seen from the small upfield shifts of the guest signals (~ 0.5 ppm, Figure S30). The resonances of both the cage and Chal guests show slight variations along the titration process, indicating a weaker shielding effect between MOC-68-Cl and Chal. Furthermore, the signals corresponding to aggregated Chal molecules in bulk solution start to appear at ~ 6 equiv Chal (3 mM), implying that the relaxed conformation of MOC-68-Cl allows a higher freedom of guests to undergo faster exchange, which is significantly different from the guest-binding behavior of the conformationally tightened MOC-68- PF_6 . Nonetheless, the solubilizing effect of MOC-68-Cl and MOC-68- PF_6 is evident. In the same $\text{CD}_3\text{CN}\text{--}\text{D}_2\text{O}$ (1:3, v/v) media, the aggregation of free Chal in blank solution occurs at 2 mM, while in MOC-68-Cl and MOC-68- PF_6 solutions, the free chalcones aggregate at 3 and 6.75 mM, respectively. This noticeable difference indicates that the MOC-68 cage in either the R-state or the T-state can trap Chal guests to increase their solubility in cage solution, albeit to distinct extents.

The fitting of the titration data with a Hill function showed a normal sigmoidal shape for MOC-68-PF₆ but an irregular profile for MOC-68-Cl, suggesting an uncertainty in comparing their binding constants for the multiple guests (Figure S31). Nevertheless, the Hill coefficient of MOC-68-PF₆ is $n = 6.24$, indicating a significantly stronger cooperative binding tendency compared to MOC-68-Cl ($n = 1.69$), implying a preferential cooperativity of MOC-68-PF₆ to allow each pocket to bind a second guest after the initial binding of the first guest molecule.

Photocatalytic Selectivity Control. The unique guest-binding behaviors mediated by anions inspired us to apply this allosteric cage in photocatalytic reactions with the expectation of regulating the product selectivity. Visible-light-photo-sensitized olefin [2 + 2] cycloadditions have attracted extensive attention over the past decades.^{47,56,57,65} However, for the intermolecular coupling reactions of acyclic olefins, the *syn*-HH diastereomers, which are considered important in many biologically active natural products, were obtained only as a minor product in most photoreactions, which is due to the fact that the *syn*-HH diastereomers are thermodynamically and sterically disfavored.^{65–67} Our group previously applied a cage catalyst of MOC-16, which possesses similar photoactive Ru-centers for sensitization and pockets suitable for guest dual stacking⁶⁸ analogous with MOC-68, to achieve the *syn*-HH diastereoselectivity of [2 + 2] cycloaddition for acyclic olefins.^{44,45} However, the rigid cage structure somewhat limits its adaptability to certain substrates for satisfactory selectivity; for example, the Chal derivatives bearing α -benzoyl functional groups generally gave poorer d.r. values as compared to those functionalized in β -position by different phenyl groups.⁴⁴ To our delight, the present allosteric cage is able to solve this problem, giving much better diastereoselectivity to this kind of reluctant substrates. Moreover, MOC-68 can act as an adjustable supramolecular reactor to tune the catalytic efficiency and selectivity of the [2 + 2] cycloaddition of chalcones artificially by anions.

To verify the influence of cage allostery on catalytic performance, the reaction conditions of [2 + 2] photocycloaddition of Chal were optimized under similar solvent systems as in NMR titration experiments for clarity (Figure 5 and Table S3). Due to the favorable solubility of both cages in the mixed organic-water solutions, a CH₃CN-H₂O mixture in 1:3 (v/v) ratio was selected as the reaction media based on the optimal yield and diastereoselectivity. In general, MOC-68-PF₆ exhibited better catalytic performance in terms of yield and d.r. value than MOC-68-Cl (Figures 6 and S34–S49 and Tables S3–S5). The Chal coupling reached a moderate yield (66%) and diastereoselectivity (d.r. = 2.7:1) for the *syn*-HH product with MOC-68-Cl as catalyst, nearly 2-fold enhanced compared to our previous result catalyzed by the rigid MOC-16⁴⁴ (Figure 6b and Table S4). Notably, the d.r. value was significantly improved by using MOC-68-PF₆ as catalyst, reaching 8.3:1, with a better yield of 71%. This means the T-state of the MOC-68 cage is superior in diastereoselectivity over the R-state, providing a more efficient photocatalytic model for the *syn*-product from acyclic olefin cycloaddition. We speculate this is because the restricted cage in MOC-68-PF₆ has more compressed open pockets to prearrange two planar aromatic Chal substrates, while the relaxed cage in MOC-68-Cl imposes a higher degree of freedom of Chal guests and is thus not so effective in the control of diastereoselectivity. It is worth mentioning that, since the 6 open pockets in the flexible tetrahedral MOC-68 cage have a longer space between two

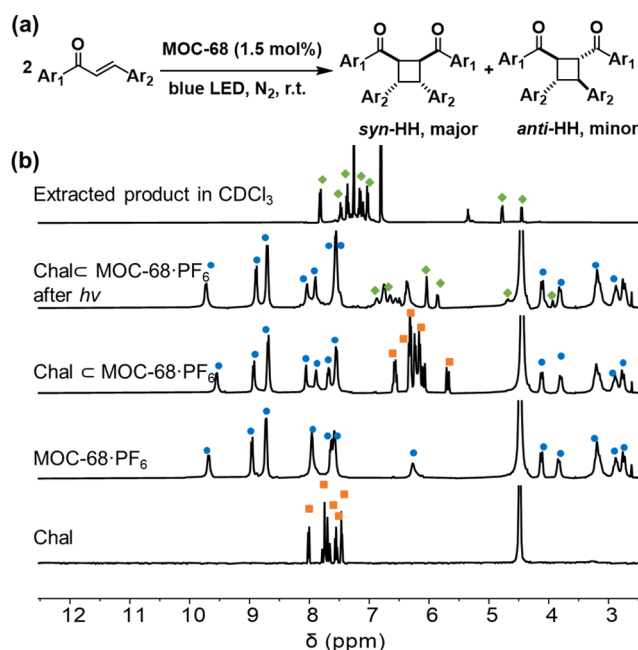


Figure 5. (a) Cage-catalyzed [2 + 2] cycloadditions of Chal derivatives. (b) ¹H NMR monitoring of Chal photocycloaddition process catalyzed with MOC-68-PF₆ (400 MHz, CD₃CN:D₂O = 1:3, v/v, 298 K). Chal is denoted by orange squares, MOC-68 by blue dots, and the *syn*-HH product by the green rhombus.

bent walls than the 12 open pockets in the rigid rhombic dodecahedral MOC-16,⁴⁴ the tolerance of the functionalizing groups on the two sides of Chal molecule will be improved, which facilitates expansion of the substrate scope with desired selectivity.

Chalcone derivatives with various mono- and bifunctional groups have been testified to show moderate to good yields for the [2 + 2] photocycloadditions catalyzed by MOC-68-PF₆ (Figure 6a and Table S4). All of the substrates produced *syn*-HH diastereomers as the major product, and better diastereoselectivities as compared with the values obtained by our previously reported rigid MOC-16 cage in the optimized condition in this work and the previously reported conditions.⁴⁴ MOC-68-Cl also generally gave improved diastereoselectivity compared to MOC-16, but the conversion efficiency is poor in most cases. Furthermore, the MOC-68-BF₄ cage with BF₄[−] as counteranion was also checked for the photocatalysis, which displayed a comparable catalytic behavior to MOC-68-PF₆ in a T-state conformation. These results highlight the significance of the allosteric effect in catalysis, implying the promise of an artificial allosteric strategy in the control of the catalytic performance.

To understand the role of MOC-68 in the [2 + 2] photocycloaddition, an in situ NMR study was performed to track the photoreaction process of Chal coupling (Figure 5b). After 450 nm LED irradiation at room temperature in N₂ for 12 h, the proton signals of the cage-binding substrates disappeared and one set of new signals attributable to the product emerged. Owing to the mismatch between the cage host and the nonplanar product after coupling, the dimerized products tend to leave the cage pockets to precipitate from the reaction media, which diminishes the product resonances on the NMR spectrum. Therefore, the product was extracted and identified by ¹H NMR, confirming the production of the major

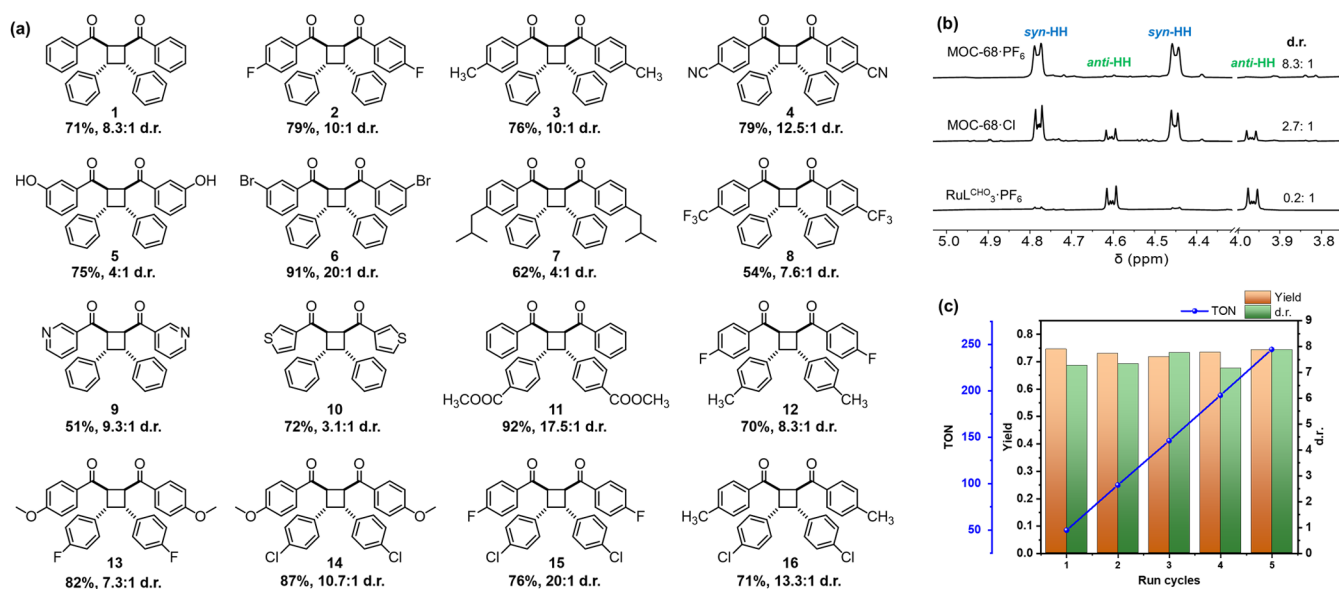


Figure 6. (a) Substrates scope of MOC-68-PF₆-catalyzed [2 + 2] cycloaddition for chalcone derivatives. (b) Comparison of *syn*- and *anti*-product selectivity based on ¹H NMR spectra (400 MHz, CDCl₃, 298 K) by using MOC-68-PF₆, MOC-68-Cl, and subcomponent RuL₃CHO₃-PF₆ as catalysts, showing d.r. values from the Chal reactant. (c) Recycling performance of MOC-68-PF₆-catalyzed Chal photocycloaddition for five cycles, showing the yields, d.r. values and TONs for recycled reactions.

syn-HH diastereomer and the *anti*-HH byproduct as the minor outcome (d.r. = 8.3:1, Figures 6b and S34). The control experiment with subcomponent RuL₃CHO₃-PF₆ as a catalyst was also carried out (Figure 6b and Table S5). As expected, the stereochemical outcome of RuL₃CHO₃-PF₆ is usual as for other molecular photocatalysts,^{65,69,70} preferring *anti*-HH products due to the lack of cage effect to control preorganization of the substrates in the confined cage space.⁷⁰ Therefore, a mechanistic model for cage-catalyzed reactions is proposed in Figure S50. Initially, two planar substrates are prearranged within a pocket of MOC-68 owing to adaptive host–guest interactions and cooperative binding, where the dynamic guest exchange and motion is subject to the spatial confinement of the cage pocket. Upon light irradiation, photosensitization proceeds through a triplet energy-transfer process from the excited state of MOC-68 to the Chal guest, leading to the formation of a diradical intermediate.⁴⁴ Subsequently, formation of a [2 + 2] cycloadduct occurs, which is stereocontrolled by the binding pocket. After dimerization, the nonplanar product becomes incompatible with the pocket microenvironment, thereby enabling its release to promote catalytic turnover of the cage reactor.

The kinetic study was further implemented by monitoring the consumption of the Chal reactant to reveal the catalytic efficiency (Figure S51). Approximate first-order fitting offered the reaction rate constants of 2.37×10^{-3} , 4.95×10^{-3} , and $1.62 \times 10^{-2} \text{ min}^{-1}$ for RuL₃CHO₃-PF₆, MOC-68-Cl, and MOC-68-PF₆ photocatalysts, respectively. These results reveal a rate enhancement effect of the cage catalysts. MOC-68-Cl catalyzes the intermolecular [2 + 2] coupling with the $k_{\text{MOC-68-Cl}}/k_{\text{Ru-L-CHO}} = 2.1$, just doubling the photoreaction due to the inferior guest encapsulation action, whereas the tightened MOC-68-PF₆ affords nearly 7-fold rate enhancement effect ($k_{\text{MOC-68-PF}_6}/k_{\text{Ru-L-CHO}} = 6.8$), which is more than 3-fold faster than that catalyzed by MOC-68-Cl. This reaction acceleration is reminiscent of enzymatic catalysis, manifesting an allosteric catalysis character of MOC-68 that is regulatable by the anions. It should be kept in mind that, although the enzyme-mimicking

cage catalysis dominates and promotes [2 + 2] cycloadditions, the concurrent photoreactions outside the cage are inevitable due to guest exchange; therefore, a deviation of the diastereoselectivity could not be avoided to deteriorate high efficiency.

Finally, we checked the recyclability and durability of the cage catalyst MOC-68-PF₆. As seen from Figure 6c, the recycling performance of the cage is prominent, showing no obvious decrease in either yield or diastereoselectivity after 5 reaction cycles. The accumulated reaction turnover (TON) numbers increase almost linearly up to ~250 at the end of the fifth round. The ¹H NMR test also confirmed the stability of the cage catalyst after repeating reactions, where no structure devastation was observed (Figure S52).

CONCLUSIONS

We have demonstrated the significance of cage allostery in boosting the enzymatic catalytic performance. The flexible feature of MOC-68 enables conformational change that can be effectively manipulated by anions. The coeffectors PF₆[−] can trigger constraint of the cage, enhancing guest-binding affinity to control their prearrangement in the cage pockets. As a consequence, the enhanced diastereoselectivity of intermolecular [2 + 2] cycloaddition is achievable for acyclic Chal derivatives, which was hardly attainable previously. The enzyme-mimicking rate enhancement effect is observed for this artificial allosteric catalysis, and MOC-68-PF₆ also exhibits excellent water solubility and stability, making it a promising candidate for green and recycling photocatalysis. This study not only underscores the role of artificial allostery in manipulating the functions of synthetic MOCs but also proves the potential of allosteric control in improving catalytic efficiency and selectivity, providing a clue to design supramolecular cages for a high level of functionalities and applications.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.3c04593>.

Materials and methods; spectral characterization; supplemental figures and tables (PDF)

Single-crystal structure crystallography, file 1 (CIF)

Single-crystal structure crystallography, file 2 (CIF)

Single-crystal structure crystallography, file 3 (CIF)

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Notes

The authors declare no competing financial interest.

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